Q #13 **Multiple Select Type** **Award: 2** **Penalty: 0** **Machine Learning**

We have data $\{(x_1, y_1), \ldots, (x_n, y_n)\}$ and fit the simple linear model

$$y = w_0 + w_1 x.$$

The design matrix is

$$X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \in \mathbb{R}^{n\times 2}, \quad w = \begin{bmatrix} w_0 \\ w_1 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}, \quad X^T X = \begin{bmatrix} 4 & 2 \\ 2 & 5 \end{bmatrix}.$$

Which of the following statement(s) is/are correct?

A. The sample size is $n = 4$.

B. $X^T y = \begin{bmatrix} 18 \\ 29 \end{bmatrix}$.

C. The sample size is $n = 2$.

D. $X^T y$ cannot be determined from the given information.

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Let $X \in \mathbb{R}^{n\times(p+1)}$ (first column all ones for the intercept; remaining p columns are features) and $y \in \mathbb{R}^n$. Consider the two ridge estimators:

$$\hat{\theta}_A = \arg\min_{\theta} \frac{1}{n}\sum_{i=1}^{n}\left(y_i - x_i^\top\theta\right)^2 + \lambda\sum_{k=0}^{p}\theta_k^2 \quad \text{(intercept penalized)}$$

$$\hat{\theta}_B = \arg\min_{\theta} \frac{1}{n}\sum_{i=1}^{n}\left(y_i - x_i^\top\theta\right)^2 + \lambda\sum_{k=1}^{p}\theta_k^2 \quad \text{(intercept not penalized)}$$

As $\lambda \to \infty$, the predictions from these models approach which limits?

A. $\hat{\theta}_A \to 0, \quad \hat{\theta}_B \to 0$

B. $\hat{\theta}_A \to 0, \quad \hat{\theta}_B \to \bar{y}$ (the sample mean of the true y values)

C. $\hat{\theta}_A \to \bar{y}, \quad \hat{\theta}_B \to 0$

D. $\hat{\theta}_A \to \bar{y}, \quad \hat{\theta}_B \to \bar{y}$

Q #15 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

Let $X \in \mathbb{R}^{n\times d}$ and $y \in \mathbb{R}^n$. In ordinary least squares linear regression, we define

$$\hat{w} = \left(X^\top X\right)^{-1}X^\top y, \quad \hat{y} = X\hat{w}, \quad r = y - \hat{y}, \quad H = X\left(X^\top X\right)^{-1}X^\top$$

Here, $\hat{y}$ is the vector of predictions, r is the residual vector, and H is known as the projection matrix .
Which of the following statements is/are true?

A. $Hy = \hat{y}$

B. $H\hat{y} = \hat{y}$

C. $Hy = r$

D. $H\hat{y} = 0$